MIL-C-47071(MI)
10 May 1974
SUPERSEDING
MIS 14194
30 June 1964

MILITARY SPECIFICATION

COATING, ABLATIVE, PHENOLIC-EPOXY RESIN SYSTEM

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a two component ablative coating.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein.

SPECIFICATION

Federal

QQ-A-362	Aluminum Alloy, Plate and Sheet

STANDARDS

Federal

FED-STD-175	Adhesive, Methods of Testing
FED-STD-406	Plastics, Methods of Testing

FSC 8030

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:42Z
Check the source to verify that this is the current version before use.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample (4.2) of material shall meet the requirements of this specification.

3.2 Material. The material shall consist of a two component, low density, filled, phenolic-epoxy resin formulation.

3.3 Mechanical properties. The material shall meet the requirements specified in Table I at 25 degrees Celsius (C) (77 degrees Fahrenheit (F)).

Table I

Mechanical Properties		
Property	Value (Minimum)	Applicable Tests
Tensile Strength Pounds per Square Inch (psi)	3,500	4.6.1.1
Modulus of Elasticity (psi)	4.25×10^5	4.6.1.2
Shear Strength, 1/2 Inch Lap (psi)	1,800	4.6.1.3

3.4 Physical properties.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:42Z
Check the source to verify that this is the current version before use.

3.4.1 Specific gravity. The specific gravity of the cured resin system shall be 0.9 to 1.2 at 25 degrees C (77 degrees F).

3.4.2 Pot life. The pot life of the uncured resin system shall be 30 minutes minimum at 25 degrees C (77 degrees F).

3.4.3 Heat resistance. The weight loss of the cured material shall not exceed 30 percent after exposure to 3093.5 degrees C (5600 degrees F) for 50 to 60 seconds.

3.4.4 Shelf life. The shelf life of the resin and hardener in unopened containers shall be 3 months minimum from date of manufacture when stored at temperatures less than 27 degrees C (80.6 degrees F).

3.5 Workmanship. The workmanship shall be such as to insure a product which is uniform and in conformance with this specification. The resin and activator shall be free of dirt, foreign material or other contaminents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of material. The preproduction sample shall be subjected to all examinations and tests specified herein. Unless otherwise specified (see 6.2), the Government will perform the examinations and tests for preproduction sample acceptance at the contractor's plant. Preproduction samples which do not meet all the requirements of this specification shall be rejected and returned to the contractor. Subsequent quantities will not be considered for acceptance until approval of the preproduction sample has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:42Z
Check the source to verify that this is the current version before use.

4.3 Classification of examinations and tests. The examination and testing of material shall be classified as follows:

a. Preproduction tests (4.4).

b. Quality conformance tests (4.5).

4.4 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

4.5 Quality conformance tests. Quality conformance tests for acceptance of material shall consist of the following examinations and tests:

a. Shear strength (4.6.1.3).

b. Specific gravity (4.6.2.1).

4.5.1 Lot size and sampling.

4.5.1.1 Lot size. Lot size shall consist of all the material submitted for acceptance at the same time, which has been prepared by the same company without change in materials or processes in one continuous period of operation.

4.5.1.2 Sampling. Unless otherwise specified, sufficient quantities to conduct the tests specified (see 4.6.1.1 and 4.6.2.1) from one unit representative of each lot shall be selected at random for quality conformance testing. Each container in a lot shall be considered as a unit of product. Failure of sample to meet the quality conformance test requirements shall be cause for lot rejection (see 4.7).

4.6 Test methods.

4.6.1 Mechanical property tests.

4.6.1.1 Tensile strength. The tensile strength specified in Table I shall be determined in accordance with FED-STD-406, Method 1011.

4.6.1.2 Modulus of elasticity. The modulus of elasticity specified in Table I shall be determined in accordance with FED-STD-406, Method 1031.

4.6.1.3 Shear strength. The shear strength tests shall be determined in accordance with FED-STD-175, Method 1033. The adherends shall be aluminum conforming to Federal Specification QQ-A-362.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:42Z
Check the source to verify that this is the current version before use.

The glue line thickness shall be 0.015 plus or minus 0.005 inch and shall be maintained under contact pressure of less than 10 pounds per square inch throughout the manufacturer's recommended cure cycle.

4.6.2 Physical property tests.

4.6.2.1 Specific gravity. The specific gravity of the cured adhesive shall be determined in accordance with FED-STD-406, Method 5011.

4.6.2.2 Pot life. The pot life shall be determined in the following manner: Weigh approximately 200 grams of resin and hardener in the mixing ratio specified by the manufacturer into a shallow tray (approximately 1/2 inch deep). Mix the ingredients thoroughly with nonabsorbent spatula or suitable mixing implement, stir frequently and check for the first signs of setting-up. The pot life shall be considered ended when the compound cannot be spread evenly and smoothly with the mixing implement. Pot life (minutes) is the elapsed time from the weighing of the ingredients until the compound fails to spread evenly and smoothly.

4.6.2.3 Heat resistance.

4.5.2.3.1 Test specimen preparation. Cast and fully cure a specimen approximately 0.800 inch in diameter by 1.000 inch in length in accordance with manufacturer's recommended cure cycle.

4.5.2.3.2 Test procedure. Weigh specimen and record weight to the nearest 0.01 gram. Apply heat to the specimen by means of an oxygen-acetylene torch adjusted to a neutral flame, or by any other suitable heating device, for 50 to 60 seconds at 3093.5 plus or minus 150 degrees C (5600 plus or minus 270 degrees F). The weight of the specimen shall be determined after heat application and the weight loss calculated for compliance with 3.4.3.

4.6.2.4 Shelf life. The material shall be tested for conformance with all the requirements specified herein after completion of the storage period. The manufacturer's certificate of compliance with the shelf life requirement (see 3.4.4) may be accepted during the storage period.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:42Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified by the procuring activity, the resin and hardener may be furnished in tin plate cans, pails or in glass jars with metal lids containing an innerseal. The size of container shall be specified by the procuring activity. The resin and hardener shall be prepared for shipment in such manner that damage during shipment and storage is prevented.

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use as an ablative coating for thermal insulation up to 3093.5 degrees C (5600 degrees F).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Whether a preproduction sample is required (see 3.1).

c. Where preproduction sample test shall be performed (see 4.2).

d. Sampling plan if other than specified (see 4.5.1.2).

e. Any special marking requirements.

f. Container size (see 5.1).

6.3 Supersession data. This specification includes the requirements of MIS 14194, dated 30 June 1964.

Custodian:
Army-MI

Preparing Activity:
Army-MI
Project No. 8030-A055

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1402

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:42Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S. MAIL

DOD-314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Missile Command
ATTN: AMSMI-RCS
Redstone Arsenal, AL 35809

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:42Z
Check the source to verify that this is the current version before use.